

- 1.) A Shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them. Wolf approaching the flock and calls out, "Wolf!" the villagers refuse to be fooled. The town goes hungry. Panic ensues.

Act \ Pre	Y	N
Y	TP	FN
N	FP	TN

- TP - Real Wolf came, warning also given
- FP - No Wolf false warning given
- FN - No Wolf will come, Wolf came, predicted it won't come.
- TN - No Wolf, no false alarm

- 2.) Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 5 of 70.

Act \ Pre	Yes	No
Cat (30)	25	5
No Cat (70)	20	50

• Total image = 100 and 70.

• Actual cat images = 30

• Actual no-cat images = 70

precision : $\frac{TP}{TP + FP} = \frac{25}{25 + 5} = 0.833$

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$= \frac{25}{25 + 20} = \frac{25}{45} = 0.5556$$

Recall = 55.56%

F₁ Score

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= 0.6667$$

F₁ Score = 66.67%

3. An email service provider wants to classify emails as spam or not spam using features.

Contains "offer" = (yes/No). Contains "free" (yes/No), e-mail/length (short/Not short).

offer	Free	length	class
Yes	Yes	short	spam
Yes	NO	long	Not spam
NO	Yes	short	spam
NO	NO	long	Not spam
Yes	Yes	long	spam
NO	NO	short	Not spam

1.) Calculate prior probabilities: $p(\text{spam})$, $p(\text{Not spam})$
Likelihood probabilities for each features.

Prior probabilities

$$P(\text{spam}) = \frac{3}{6} = 0.5$$

$$P(\text{Not spam}) = \frac{3}{6} = 0.5$$

Conditional probabilities

Feature	count	probability
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offer = yes	2	$2/3$
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offer = NO	1	$1/3$
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Free = yes	3	$3/3 = 1$
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Free = NO	0	$0/3 = 0$
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length = short	2	$2/3$
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length = long	1	$1/3$
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Not spam

Feature	Count	probability
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offer = yes	1	$1/3$
-------------	---	-------

offer = NO	2	$2/3$
------------	---	-------

Free = yes	0	$0/3 = 0$
------------	---	-----------

Free = NO	3	$3/3 = 1$
-----------	---	-----------

length = short	1	$1/3$
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length = long	2	$2/3$
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3. Classify the new Email

$x = (\text{Offer} = \text{yes}, \text{Free} = \text{NO}, \text{length} = \text{short})$

For Naive Bayes:

$$P(\text{spam} | x) \propto p(\text{spam}) p(\text{Offer} = \text{yes} | \text{spam})$$

Spam score

$$P(\text{spam} | x) \propto \frac{1}{2} \times \frac{2}{3} \times 0 \times \frac{2}{3}$$

$$= 0$$

Non Spam Score

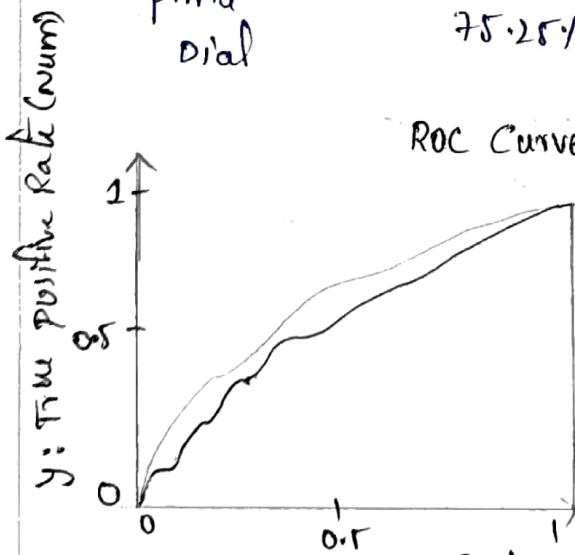
$$P(\text{Not spam} | x) \propto p(\text{Not spam}) p(\text{Offer} = \text{yes})$$

$$= \frac{1}{2} \times \frac{1}{3} \times 1 \times \frac{1}{3} = \frac{1}{18} = 0.0556$$

Since,
 $0.0556 > 0$

Dataset	Bayes Net	Naive Bayes	Naive Bayes Multinomial
Iris	93.20%	98.53%	95.92%
pima diab	75.25%	75.75%	59.82%

ROC Curve - pima - diabetes



— Naive Bayes
(tested negative)

— Random
forest
(tested negative)